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Contribution of biopsychosocial risk factors to nonspecific neck pain in office workers: A path analysis model

Arpalak PAKSAICHOL[1], Chaipat LAWSIRIRAT[2] and Prawit JANWANTANAKUL[1]

[1]Department of Physical Therapy, Faculty of Allied Health Sciences, Chulalongkorn University, Thailand and [2]Faculty of Sports Science, Chulalongkorn University, Thailand

Abstract: Contribution of biopsychosocial risk factors to nonspecific neck pain in office workers: A path analysis model: Arpalak PAKSAICHOL, *et al*. Department of Physical Therapy, Faculty of Allied Health Sciences, Chulalongkorn University, Thailand—Objective: The etiology of nonspecific neck pain is widely accepted to be multifactorial. Each risk factor has not only direct effects on neck pain but may also exert effects indirectly through other risk factors. This study aimed to test this hypothesized model in office workers. **Methods:** A one-year prospective cohort study of 559 healthy office workers was conducted. At baseline, a self-administered questionnaire and standardized physical examination were employed to gather biopsychosocial data. Follow-up data were collected every month for the incidence of neck pain. A regression model was built to analyze factors predicting the onset of neck pain. Path analysis was performed to examine direct and indirect associations between identified risk factors and neck pain. **Results:** The onset of neck pain was predicted by female gender, having a history of neck pain, monitor position not being level with the eyes, and frequently perceived muscular tension, of which perceived muscular tension was the strongest effector on the onset of neck pain. Gender, history of neck pain, and monitor height had indirect effects on neck pain that were mediated through perceived muscular tension. History of neck pain was the most influential effector on perceived muscular tension. **Conclusions:** The results of this study support the hypothesis that each risk factors may contribute to the development of neck pain both directly and indirectly. The combination of risk factors necessary to cause neck pain is likely occupation specific. Perceived muscular tension is hypothesized to be an early sign of musculoskeletal symptoms.
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Key words: Biopsychosocial, Neck pain, Office worker, Path analysis, Predictors

Neck pain is common among workers, affecting 13–48% of workers annually[1,2]. Office workers, defined as those working in an office environment with their main tasks involving computer use, participation in meetings, presentations, reading, and making and receiving telephone calls[3], are among those with the highest frequency of neck pain[4]. Previous studies showed that between 42 and 69% of office workers experienced neck pain in the preceding 12 months[5–8] and that about 34 to 49% of office workers reported a new onset of neck pain during a one-year follow-up[9–11]. Neck pain is viewed as an episodic occurrence over a lifetime with variable recovery between episodes[12]. It has been found to increase the risk for future long-term sickness absence among white-collar workers[13]. Consequently, neck pain in effect constitutes a great socioeconomic burden on patients and society[14, 15].

Nonspecific neck pain is neck pain (with or without radiation) without any specific systematic disease being detected as the underlying cause of the complaint[16]. The etiology of musculoskeletal disorders is widely accepted to be multifactorial, including individual, physical, and psychosocial factors[14, 17–19]. Different occupations are exposed to different working conditions, and the nature of the work influences the health of workers[14, 20–22]. Predisposing factors for neck pain are likely to be population specific. A recent systematic review of prospective cohort studies identified several risk factors for neck pain in office workers, including female gender, history of neck complaints, pain initially felt after an accident, irregular head and body posture, duration of employment in same job <1 year (for males only), poor computer

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Correspondence to: P. Janwantanakul, Department of Physical Therapy, Faculty of Allied Health Sciences, Chulalongkorn University, Bangkok, 10330, Thailand (e-mail: prawit.j@chula.ac.th)
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skills (for males only), distance of the keyboard from the edge of the table <15 cm, high task difficulty, low influence at work (for female subjects only) and high muscular tension[17].

To understand the etiology of neck pain, a model to conceptualize the process involved in the development of neck pain among workers is required. Côté *et al.*[14] proposed that, rather than each risk factor only having direct effects on neck pain and the risk factors not themselves being outcomes of antecedent risk factors, neck pain is likely caused by multiple serial exposures. For example, several risk factors relating to demographic, ethnic, and cultural characteristics may have direct effects on neck pain. However, they may also exert their effects indirectly through health behaviors, occupation, workplace physical and psychological exposures and how a worker copes with stress at work. On the other hand, risk factors related to the workplace can modify the direct effects of other workplace-related risk factors on the onset of neck pain, and the effects of risk factors related to the workplace on neck pain are mediated by workers' ability to cope with stress at work.

An analytical approach, such as path analysis, is a useful tool to test a causal pathway for the development of disease. Path analysis, which is an extension of multiple regression, can predict more than one dependent variable and assess the relationships among independent variables as well as dependent variables within that model[23, 24]. It shows a theoretical, directional relationship (both direct and indirect) between variables, and offers a causal model of relationships[25]. The purpose of this study was to test a hypothesized model of the direct and indirect effects of various risk factors involved in the development of nonspecific neck pain in a sample of office workers using path analysis.

Materials and Methods

Participants and procedure

A prospective cohort study with a one-year follow up was conducted in a convenience sample of 3,809 office workers recruited from four large-scale workplaces in Bangkok, Thailand. The enterprises participating in this study were a public university and the head offices of three ministries. The study was approved by the Chulalongkorn University Human Ethics Committee. Individuals were included in the study if they were 18–55 years of age and working full-time. Subjects were excluded if they had reported neck pain in the previous three months, reported pregnancy or had planned to become pregnant in the next 12 months, had a history of trauma or accidents or surgery in the neck region, had been diagnosed with fibromyalgia, carpal tunnel syndrome, cervical radiculopathy, systemic illness or connective tissue disorders, or planned to be on vacation more than 9 consecutive days in the next 12 months. Neck pain was defined as any neck pain lasting >24 hours with a pain intensity greater than 30 mm on a 100-mm visual analog scale (VAS).

Office workers were approached and invited to participate in this study. They were informed of the objectives and details of the research and asked to provide informed consent upon agreement to participate. At baseline, subjects completed a self-administered questionnaire and underwent a physical examination conducted by trained physical therapists according to a standardized protocol (Fig. 1). Subjects then received a self-administrated diary to record the incidence of neck pain monthly, and the researcher returned to collect the diaries from the participants every month over a 12-month period.

Outcome measures

To assess onset of neck pain during the previous month, a picture of the body from the standardized Nordic questionnaire[26] and the question "Have you experienced any neck pain lasting >24 hours in the previous four weeks?" were included in the diary

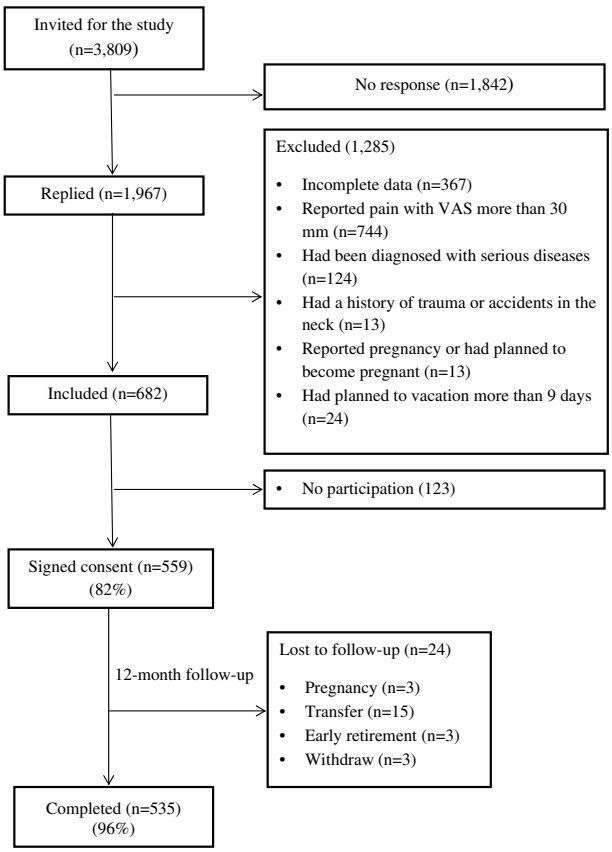


Fig. 1. Flow chart of participants.

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given to the participants. If they answered "Yes", follow-up questions were asked regarding pain intensity measured by a VAS and the presence of weakness or numbness in the upper limbs. In this study, participants were identified as cases if they answered "Yes" to the question "Have you experienced any neck pain lasting >24 hours in the previous four weeks?" reported a pain intensity greater than 30 millimeters (mm) on a 100-mm VAS and had no weakness or numbness in the upper limbs. Participants were followed until they became symptomatic, withdrew from the study or completed the 12-month follow-up.

Biopsychosocial risk factors

A self-administered questionnaire and physical examination were employed to collect potential biopsychosocial risk factors. The self-administered questionnaire consisted of three sections in order to gather data on individual, work-related physical, and work-related psychosocial factors.

Individual factors included gender, age, hand dominance, marital status, education level, chronic diseases, frequency of weekly exercise sessions, smoking habits, and history of neck and low back pain.

Work-related physical factors included job position, years of working experience, average number of working hours a day and frequency of computer use and sitting >4 h a day as well as rest breaks. Information about typing style and habitual neck posture while using a computer was also requested. The questionnaire asked participants, based on their own perceptions, to rate the ergonomics of their workstations (i.e., height of their desk and chair, adjustability of their chair and positions of their computer screen, keyboard and mouse) and work environment conditions (i.e., ambient temperature, light intensity, noise level, air circulation).

Work-related psychosocial factors were assessed with the Job Content Questionnaire, which consists of a set questions covering a total of 54-items in the following six areas: decision latitude (11 items), psychological demand (12 items), physical job demand (6 items), social support (8 items), job security (5 items) and work hazards (12 items). Each item had a response set of a four-point Likert scale ranging from 1, strongly disagree, to 4, strongly agree[27]. Participants were also required to answer the question "Have you, during the past month, experienced muscular tension during work?" (never, a few times, a few times per week, one time per day or several times per day). Self-rated perceived muscular tension was scaled into three groups: high tension (a few times per week, one time per day or several times per day), medium tension (a few times) or low tension (never)[11].

The physical examination included in the study was selected based on the theoretical effect of prolonged computer use on body parts, which may lead to forward head posture, rounded shoulders and kyphotic upper thoracic spine[28]. Previous studies showed that patients with neck pain had significantly lower ranges of neck movement and neck muscle endurance than those without neck pain[5,29]. The physical examination took a single 30-minute session to complete.

- Body weight and height were measured by digital scale and a wall-mounted standiometer, respectively.
- Neck range of motion assessments looked at the active range of motion for neck flexion, extension, rotation and lateral rotation using the cervical range of motion device (CROM)[30]. Subjects sat on a chair with their feet on the floor. Each subject looked directly forward with his/her neck in a neutral position. The subject was then asked to move his/her head towards each direction as far as possible, and the degree of neck motion in each direction was recorded.
- Neck extensor and flexor endurance were assessed according to the procedures described by Ljungquist *et al.*[31] and Harris *et al.*[32], respectively. For the neck extensor muscles endurance, the subject lay prone on a plinth with their head and neck supported by the examiner's hands. A Velcro band was strapped around the subject's head with an inclinometer attached to the band immediately above the tip of the right ear. A 2-kg weight (for females) or 4-kg weight (for males) was suspended from the headband. The subject was instructed to hold his/her the head steady in a horizontal position, and this was monitored with an inclinometer. The test was discontinued if the subject was not able to hold the position because of fatigue or pain or if the subject moved >5 degrees from the position. The examiner recorded the muscle performance in seconds[31]. For neck flexor muscle endurance, the participant assumed a crook-lying position with their chin maximally retracted and maintained isometrically. The subject then lifted his/her head and neck until the head was approximately 2.5 cm above the plinth. The length of time the subject was able to hold this position without deviation was recorded in seconds by the examiner[32].
- Pressure pain threshold (PPT), which is the minimal amount of pressure where the sensation of pressure first changes to pain, at the right upper trapezius was measured using an electronic algometer[33]. The pressure was applied at a rate of 30 kPa/s. All participants were instructed to press a switch when the sensation changed from pressure to pain. The mean of three trials was calculated and used for the main analysis. A 30-second resting period was allowed between each measurement.

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Before data collection, the repeatability of data from the self-administered questionnaire and physical examination outcomes was assessed on 20 office workers. Each subject was tested twice on two separate days with a week between the measurements for the questionnaire and one day for the physical examination.

Statistical analyses

For the reliability study, the intraclass correlation coefficient (ICC [3,1]) was calculated for continuous data and Phi coefficient was calculated for nominal data. The ICC (3,1) was calculated for intra-rater reliability.

Characteristics of subjects were described using means or proportions. The percentages of missing data for the individual, work-related physical and work-related psychosocial factor categories were 0.1%, 1.1% and 0.5%, respectively. To retain the statistical power of the database, missing data were handled by the "hot-deck imputation" procedure. A respondent was selected at random from the total sample of the study, and the value for that person was assigned to the case in which information was missing. This procedure was conducted repeatedly for each missing value until the dataset was complete[34]. The baseline and 12-month follow-up characteristics of the study population were compared using Chi-square analysis and the independent *t*-test.

Descriptive analysis and multiple regression were performed using SPSS for Windows Version 17.0, and path analysis was performed using LISREL Version 8.5. To test the hypothesized model, a three-step process analysis was undertaken. First, univariate logistic regression analysis was carried out to determine significant differences in the onset of neck pain with various biopsychosocial characteristics. Any factors with a p-value ≤0.2 in the univariate logistic regression analysis were eligible for addition into multivariate analysis. Second, multivariate logistic regression was conducted to determine whether baseline measures of biopsychosocial risk factors were associated with incident neck pain. Third, path analysis was used to examine the relationships among various factors with incident neck pain based on the model proposed by Côté *et al.*[14]. The overall model fit was assessed by establishing fit indexes: the chi-square significance test (χ^2), the root mean square error of approximation (RMSEA), the comparative fit index (CFI) and the goodness-of-fit index (GFI). The χ^2 statistic was used where a nonsignificant test indicates that the model and data were consistent. The RMSEA is an index of the amount of mis-specification of the model per degree of freedom, where values less than 0.05 indicate a good fit, values between of 0.05 and 0.08 indicate a marginal fit, and values greater than 0.1 indicate an unacceptable fit. Fit index values more than 0.95 are considered to indicate acceptable fit of a model to data[35].

Results

Test-retest reliability

The test-retest reliability results demonstrated moderate (0.71) to good (0.91) reliability for self-administered questionnaire outcomes. Intra-rater reliability for physical examination outcomes were moderate (0.72) to good (0.91).

Demographic characteristics of study population

Of the total 3,809 office workers who received a letter inviting them to participate in the study, 1,967 responded (52%). Of these, 682 were eligible and 559 agreed to participate at baseline measurement. A total of 535 office workers were followed for 1 year, and 24 (5%) subjects were lost during the follow-up period due to pregnancy (n=3), job transfer (n=15), early retirement (n=3) and withdrawal (n=3). Table 1 shows the baseline and 12-month follow-up characteristics of the study population. No significant difference in subject characteristics between the baseline and 12-month follow-up data was detected (p>0.05). Over the 12 months of follow-up, 28% (151/535) of participants reported the incidence of neck pain with a mean (SD) VAS score of 42 (14) mm.

A conceptual model for the onset of nonspecific neck pain in office workers

Because there were significantly different numbers of female (n=429) and male office workers (n=106) participating in the study, multiple regression and path analysis were separately conducted for each gender. However, a model of the direct and indirect effects of various risk factors involved in the development of neck pain in female office workers was not significantly different from that of their male counterparts. Thus, data from females and males were combined for further multiple regression and path analysis.

When performing univariate logistic regression analyses, the variables showing a p-value ≤0.2 were gender, history of neck pain and back pain, neck flexor endurance, desk and monitor height, typing style, perceived muscular tension, and physical job demand. Thus, these factors were selected for further analysis. Multivariate logistic regression analyses revealed that gender, history of neck pain, monitor height, and perceived muscular tension were associated with onset neck pain (Table 2).

The causal relationships of gender, history of neck pain, monitor height and perceived muscular tension with onset of neck pain were examined by path

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Table 1. Descriptive summary of subject characteristics

Characteristics	Baseline (n=559)		12-month follow-up (n=535)		*p*
	N (%)	Mean (SD)	N (%)	Mean (SD)	
General characteristics					
Gender					0.711*
Male	113 (20.2)		106 (19.8)		
Female	446 (79.8)		429 (80.2)		
Age (years)		39.1 (9.1)		39.2 (9.0)	0.848**
20–29	105 (18.8)	26.6 (1.8)	97 (18.1)	26.6 (1.8)	0.945**
30–39	195 (34.9)	34.8 (2.8)	186 (34.8)	34.8 (1.9)	0.889**
40–49	161 (28.8)	44.2 (2.9)	157 (29.3)	44.2 (2.9)	0.959**
50–59	98 (17.5)	53 (2.5)	95 (17.8)	52.7 (1.9)	0.821**
Height		1.6 (0.1)		1.6 (0.1)	0.809**
Body weight		61.1 (14.3)		61.3 (14.4)	0.932**
Body mass index (kg/m^2)					0.747*
<18.5 kg/m^2	34 (6.1)	17.7 (0.6)	31 (5.8)	17.7 (0.6)	0.917**
18.5–24.9 kg/m^2	347 (62.1)	21.7 (1.7)	331 (61.9)	21.7 (1.7)	0.984**
25–29.9 kg/m^2	112 (20.0)	27.2 (1.3)	110 (20.6)	27.3 (1.4)	0.890**
>30 kg/m^2	66 (11.8)	34.7 (5.4)	63 (11.8)	34.3 (3.8)	0.853**
Marital status					0.971*
Single	316 (56.5)		300 (56.1)		
Married	219 (39.2)		212 (39.6)		
Divorced/widowed/separated	24 (4.3)		23 (4.3)		
Level of education					0.928*
Primary school	8 (1.4)		8 (1.5)		
Secondary school	13 (2.3)		13 (2.4)		
College	70 (12.5)		69 (12.9)		
Bachelor's degree	372 (66.5)		354 (66.2)		
Higher than Bachelor's degree	96 (17.2)		91 (17)		
Frequency of weekly exercise sessions					0.985*
No	155 (27.7)		147 (27.5)		
Sometimes	341 (61.0)		326 (60.9)		
Always	59 (10.6)		58 (10.8)		
Not sure	4 (0.7)		4 (0.8)		
History of neck pain					0.814*
Yes	284 (50.8)		262 (49)		
No	275 (49.2)		273 (51)		
History of back pain					0.792*
Yes	352 (63)		335 (62.6)		
No	207 (37)		200 (37.4)		
History of illness					0.996*
Yes	99 (17.7)		97 (18.2)		
No	460 (82.3)		438 (81.8)		
Work-related physical characteristics					
Duration of employment (years)		13.5 (9.2)		13.8 (9.3)	0.816**
Working days per week (days per week)		5.0 (0.2)		5.0 (0.3)	0.872**
Working hours per day (hours per day)		8.0 (1.0)		8.0 (1.0)	0.989**

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Table 1. Descriptive summary of subject characteristics (Continued)

Characteristics	Baseline (n=559)		12-month follow-up (n=535)		*p*
	N (%)	Mean (SD)	N (%)	Mean (SD)	
Adjustable chair					0.963*
Yes	456 (81.6)		340 (63.6)		
No	103 (18.4)		195 (36.4)		
Monitor height at a level horizontal with the eyes					0.999*
Yes	322 (57.6)		306 (57.2)		
No	237 (42.4)		229 (42.8)		
Suitable desk height					0.958*
Yes	495 (88.6)		475 (88.8)		
No	64 (11.4)		60 (11.2)		
Typing					0.787*
Touch typing	294 (52.6)		278 (52)		
Non-touch typing	265 (47.4)		257 (48)		
Forward head posture while using a computer					0.998*
Often	210 (37.6)		195 (36.4)		
Sometimes	219 (39.2)		214 (40)		
Seldom	130 (23.3)		126 (23.6)		
Work-related psychosocial characteristics					
Job control		35.2 (5.0)		35.2 (5.0)	0.968**
Psychosocial job demand		32.6 (4.5)		32.7 (4.5)	0.973**
Physical job demand		13.5 (2.8)		13.5 (2.8)	0.987**
Job security		16.5 (1.5)		16.5 (1.5)	0.984**
Social support		30.1 (4.8)		29.7 (6.0)	0.895**
Hazard at work		16.7 (3.6)		16.7 (3.6)	0.874**
Perceived muscular tension					0.897*
Low	173 (30.9)		166 (31)		
Medium	297 (53.1)		284 (53.1)		
High	89 (15.9)		85 (15.9)		
Physical characteristics					
Cervical flexion (degrees)		61 (9)		61 (9)	0.911**
Cervical extension (degrees)		65 (11)		65 (11)	0.764**
Cervical rotation (degrees)					
Right		72 (7)		72 (7)	0.979**
Left		72 (7)		72 (7)	0.947**
Cervical lateral flexion (degrees)					
Right		42 (7)		42 (7)	0.758**
Left		44 (7)		44 (7)	0.950**
Neck flexor endurance (seconds)		30.7 (21.9)		30.2 (21.2)	0.830**
Neck extensor endurance (seconds)		177.0 (106.9)		176.6 (106.3)	0.779**
Pressure pain threshold (kPa)					
Right		299.8 (184.6)		300.0 (184.5)	0.890**
Left		258.4 (158.3)		258.0 (158.1)	0.641**

*Using the Chi-square test for a comparison between baseline and 12-month follow-up data. **Using the independent *t*-test for a comparison between baseline and 12-month follow-up data.

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Table 2. Incidence, β coefficient and adjusted odds ratio (OR_{adj}) with 95% confidence interval (95% CI) for nonspecific neck pain with respect to factors in the final modeling[a] (n=535)

Factors	N	Incidence (%)	β coefficient	OR_{adj}	95% CI	p value
Gender						
Female	429	131 (30.5)	0.536	1.71	1.01–2.95	0.05
Male	106	20 (18.9)		1.00		
History of neck pain						
Yes	262	95 (36.3)	0.493	1.64	1.08–2.49	0.02
No	273	56 (20.5)		1.00		
Perceived muscular tension						
High	85	41 (48.2)	1.317	3.73	1.98–7.03	<0.001
Medium	284	84 (29.6)	0.629	1.88	1.13–3.13	0.016
Low	166	26 (15.7)		1.00		
Monitor height at a level horizontal with the eyes						
Yes	306	74 (24.2)		1.00		
No	229	77 (33.6)	0.41	1.51	1.02–2.23	0.041

[a]Factors included in the statistical modelling were gender, history of neck pain and back pain, neck flexor endurance, desk and monitor height, typing style, perceived muscular tension, and physical job demand.

analysis (Fig. 2). All factors had a direct effect on the onset of nonspecific neck pain in office workers. The most influential factor causing neck pain was perceived muscular tension (β=0.19), followed by history of neck pain (β=0.12), gender (β=0.11), and monitor height (β=0.10), respectively. Factors mostly influencing perceived muscular tension were history of neck pain (β=0.35), followed by gender (β=0.07) and monitor height (β=0.05), respectively. Based on the fit indices of path analysis, this model provided a good fit for the data (Chi-square=0.00, p-value=1.00, RMSEA<0.001, GFI=0.985, CFI=0.993).

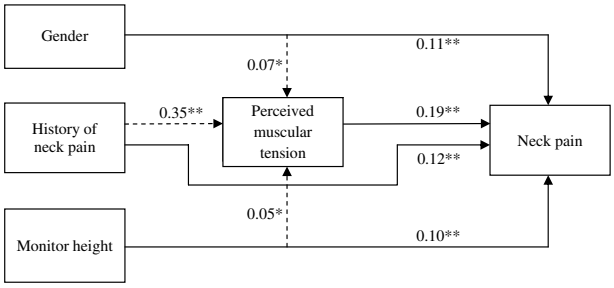


Fig. 2. Path analysis of factors predicting onset of neck pain in office workers with standardized regression coefficients (*p<0.05, **p<0.01).

Discussion

The one-year incidence of neck pain in our sample of office workers was 28%. Previous epidemiological studies reported the annual incidence of neck pain in office workers to be in the range of 34–49%[9–11]. In this study, apart from having pain lasting more than one day, participants were required to report pain greater than 30 mm on a 100-mm visual analogue scale and no weakness or numbness in the upper limbs in order to be identified as cases. Korhonen *et al.*[9] defined incident cases as those who reported local neck pain or radiating neck pain for at least eight days during the preceding 12 months, whereas Hush *et al.*[10] defined an episode of neck pain as a period of neck pain lasting longer than 24 hours. Consequently, the discrepancy between our study and previous studies may be due to the difference in the definition of a symptomatic case.

Côté *et al.*[14] suggested that most neck pain in workers is nontraumatic and that its etiology is multifaceted, meaning that neck pain is not caused by a single risk factor but rather is caused by a combination of risk factors. The specific combinations of risk factors necessary to cause an episode of neck pain likely vary between workers. The authors further elaborated about the complex relationships between individual, work-related physical and psychosocial factors for the development of neck pain by stating that each risk factor has both direct and indirect effects on the development of neck pain. Some risk factors may exert their effects indirectly through other risk factors as a mediator.

The results of the present study indicate that the onset of neck pain in office workers was predicted by gender, history of neck pain, monitor height, and perceived muscular tension. As proposed by Côté *et al.*[14], each risk factor had direct and indirect effects on the development of nonspecific neck pain in a sample

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of office workers. The model showed that female gender, having history of neck pain, monitor position not being level with the eyes and frequently perceived muscular tension directly caused neck pain and that perceived muscular tension had the strongest effect on the onset of neck pain. A recent systematic review of prospective cohort studies showed strong evidence for the history of neck complaints and female gender as risk factors of the onset of neck pain in office workers[17]. Computer screen position not being level with the eyes was also previously reported to be a predictor for the onset of neck pain in undergraduate students[36]. Several studies have reported an association between perceived muscular tension and the onset of neck pain[11, 37, 38]. Wahlström *et al.*[11] demonstrated that perceived muscular tension was significantly associated with an increased risk of developing neck pain among computer users. Huysmans *et al.*[38] found that perceived muscular tension was a strong predictor of future neck-shoulder symptoms in symptom-free office workers.

Apart from having a direct effect on the development of neck pain, gender, history of neck pain, and monitor height had indirect effects on neck pain that were mediated through perceived muscular tension. Female gender, having history of neck pain, and monitor position not being level with the eyes were related to frequently perceived muscular tension. The results also pointed out that history of neck pain had the most influential effect on perceived muscular tension.

The conceptual model for the onset of nonspecific neck pain in office workers proposed in this study is in line with an existing model of musculoskeletal disorders and computer work proposed by Wahlström[39], who hypothesized that work technology and organization have a direct path to physical demands. Both physical demands from work and mental stress may increase the physical load, which in turn has a direct path to perceived muscular tension. Individual factors are hypothesized to be effect modifiers for the association between physical demands and physical load as well as the association between work organization and mental stress. Perceived muscular tension, along with perceptions of comfort and exertion, is hypothesized to be an early sign of musculoskeletal symptoms that arises as a result of work organizational and psychosocial factors as well as physical load and individual factors. Interventions to prevent musculoskeletal disorders due to computer work should be directed at more than one factor, i.e., physical, work organizational, and psychosocial factors.

From the findings of the current study, the prevention of nonspecific neck pain among office workers should at least focus on developing strategies or interventions to rectify monitor height and to alleviate perceived muscular tension. For the other two non-modifiable risk factors (i.e., gender and history of neck complaints), this information is useful for clinicians to identify office workers at risk, which would result in enhancement of resource allocation to those most in need and most likely to benefit from it. Otherwise, a large number of people would receive interventions, which would likely compromise their effectiveness[17].

Strengths and limitations of the study

A major strength of this study is its prospective design and the evaluation of a broad range of biopsychosocial factors for their contribution to neck pain. In addition, homogenous participants, in terms of working characteristics, were selected for the present study because different occupations are exposed to different working conditions and the nature of work influences the health of workers[14]. Consequently, predisposing factors for neck pain are likely to be population-specific. However, the current study has three methodological limitations. First, in this study, subjects were identified as cases if they reported pain lasting more than 1 day, pain greater than 30 mm on a 100-mm visual analog scale and no weakness or numbness in the upper limbs. Different results may emerge with different definitions of symptomatic cases. Second, the nature of several biopsychosocial factors and the diagnosis of neck pain were subjective, which may have led to data inaccuracy. The important drawback of self-reported data is the risk of overestimation of exposure[40]. Also, some workers may be more sensitive to any somatic disturbance than others. As a result, there is a risk of under- or over-reporting of incidence. Future studies should consider the inclusion of objective information from a physical examination to increase data accuracy. Third, monitor height, which was one of the risk factors identified in this study, was categorized into only two groups; that is, the monitor was or was not positioned at a level horizontal with the eyes. Thus, the effect of a monitor height above or below eye level on the onset of neck pain could not be examined in this study. A further study should investigate this issue to enhance understanding regarding the relationship between monitor height and nonspecific neck pain in office workers.

Conclusion

We found that gender, history of neck pain, monitor height, and perceived muscular tension were predictors for nonspecific neck pain in office workers. A conceptual model for the development of neck pain was developed using path analysis. Female gender, having

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history of neck pain, monitor position not being level with the eyes, and frequently perceived muscular tension directly caused neck pain, with perceived muscular tension having the strongest effect on neck pain. Also, gender, history of neck pain, and monitor height had indirect effects on neck pain that were mediated through perceived muscular tension, with history of neck pain having the most influential effect on perceived muscular tension. Interventions aimed at preventing the occurrence of nonspecific neck pain in the office environment should address these factors.

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